Advanced Disaster Response Wi-Fi Beacon Locator: UAV-Based RSSI Localization and Identity Fingerprinting

Executive Summary

The frequency and severity of extreme weather events, particularly cyclonic storms and catastrophic flooding across the Indian subcontinent, have placed unprecedented strain on emergency management infrastructure. Historical data from disaster management operations, such as the National Disaster Response Force (NDRF) deployments during Super Cyclone Amphan and Cyclone Remal, underscore the critical necessity for rapid, high-precision search and rescue (SAR) methodologies. Traditional SAR operations rely heavily on visual identification and acoustic sensors, both of which are severely compromised by dense physical debris, extensive floodwaters, and adverse weather conditions. However, the ubiquitous proliferation of cellular devices provides a crucial, unexploited lifeline: the continuous transmission of 802.11 Wi-Fi probe requests.

This comprehensive report details the architectural design, signal processing frameworks, and algorithmic deployment of an Unmanned Aerial Vehicle (UAV) based Wi-Fi Beacon Locator. Designed to operate autonomously in post-disaster environments without reliance on active cellular infrastructure, the system leverages a heterogeneous edge computing platform to passively sniff unauthenticated Wi-Fi frames emitted by victims' devices. By coupling these radio frequency (RF) intercepts with high-resolution UAV telemetry via the MAVLink protocol, the system dynamically maps Received Signal Strength Indicator (RSSI) data to precise geospatial coordinates. The architecture overcomes severe multipath fading and signal attenuation through advanced mathematical propagation models—specifically the Log-Distance Path Loss Model for rubble environments and the Two-Ray Ground Reflection Model for flooded terrain. Furthermore, to ensure accurate distance mapping and continuous tracking despite modern MAC address randomization protocols, the payload incorporates Kalman filtering for signal smoothing and machine learning architectures for device fingerprinting and non-linear coordinate regression.

Heterogeneous Edge Computing Architecture

The computational demands of airborne SAR payloads require an intricate balance between high-throughput data processing for machine learning inferences and deterministic, low-latency execution for flight controller communication and sensor management. To achieve this, the system discards traditional single-core microcontrollers in favor of the Arduino Uno Q, a development board engineered with a "dual-brain" heterogeneous architecture.

The primary computational engine is the Qualcomm Dragonwing QRB2210 System-on-Chip (SoC), an application processor (MPU) featuring a quad-core Arm Cortex-A53 architecture clocked at 2.0 GHz. Operating on a 1.8 V logic domain, this MPU runs a Debian Linux operating system with upstream support, providing a robust environment for executing complex Python scripts, TensorFlow-based machine learning models, and extensive data logging. The SoC is supplemented by an Adreno 702 GPU and dual Image Signal Processors (ISPs), providing the necessary bandwidth for high-speed AI tasks and potential computer vision integrations. For memory management, the system utilizes 4 GB of single-rank, 32-bit LPDDR4X RAM and 32 GB of onboard eMMC storage, which is critical for buffering massive packet capture (PCAP) files without relying on failure-prone external SD cards in physically volatile environments.

Operating in parallel is a dedicated real-time microcontroller (MCU), the STMicroelectronics STM32U585, built upon an Arm Cortex-M33 architecture operating at 160 MHz. The MCU functions on a 3.3 V logic domain and executes the Arduino Core atop the Zephyr Real-Time Operating System (RTOS). This segregation of duties allows the MCU to manage precise timing operations, analog-to-digital conversions, hardware interrupts, and serial telemetry routing without being preempted by the Linux kernel's scheduling overhead. Communication between the Linux MPU and the Zephyr MCU is facilitated by a built-in Remote Procedure Call (RPC) bridge library, ensuring seamless asynchronous data transfer.

To contextualize the technical feasibility and suitability of this hardware for edge AI localization, a comparative analysis of contemporary single-board computers is provided below.

Specification	Arduino UNO Q	Raspberry Pi 5	NVIDIA Jetson Orin Nano
Processor Architecture	Dual: 4x Cortex-A53 (MPU) + Cortex-M33 (MCU)	Single: 4x Cortex-A76	Single: 6-core Cortex-A78AE
Operating System	Debian Linux + Zephyr RTOS (Dual-OS)	Raspberry Pi OS (Linux)	Ubuntu 20.04 + JetPack SDK
Wireless Connectivity	Built-in Wi-Fi 5 (Dual-band) + BT 5.1	Built-in Wi-Fi 5 + BT 5.0	Requires external dongle/module
Logic Voltage Domain	Mixed: 1.8 V (MPU) and 3.3 V (MCU)	3.3 V	Mixed: 1.8 V and 3.3 V
Primary Use Case Match	Real-time sensor control + Edge ML	General Linux computing	High-performance vision AI

Table 1: Comparative architecture analysis of edge computing platforms for UAV integration.

Power management is a critical factor for UAV payloads, where battery endurance directly dictates search coverage. The Arduino Uno Q accepts dual power inputs, supporting a 7-24V DC input through a buck converter that integrates via a diode-OR circuit onto a primary 5V system bus. This allows the payload to draw power directly from the UAV's primary lithium-polymer (LiPo) flight batteries, eliminating the weight of secondary power sources while providing reverse-polarity protection to safeguard the electronics during rapid field deployments.

UAV Telemetry and MAVLink Protocol Integration

The physical transportation of the payload is governed by an autonomous multirotor UAV operating the ArduPilot open-source flight stack, an industry standard for reliable autonomous navigation. For the spatial localization algorithms to function, every intercepted Wi-Fi packet must be instantly correlated with the exact 3D spatial coordinates of the UAV at the moment of reception.

This synchronization is achieved via MAVLink (Micro Air Vehicle Link), a highly efficient, lightweight messaging protocol designed specifically for the drone ecosystem. MAVLink operates on a modern hybrid publish-subscribe and point-to-point design, transmitting binary packets over serial connections with minimal overhead. The STM32 MCU continuously parses the incoming MAVLink data stream from the flight controller, extracting the **GLOBAL_POSITION_INT** message (providing latitude, longitude, and relative altitude) and the **ATTITUDE** message (providing pitch, roll, and yaw angles).

By offloading the serial parsing to the real-time MCU, the system avoids the latency inherent in USB or Linux-based serial buffers. Once a Wi-Fi probe request is intercepted by the Linux MPU, it issues an RPC call to the MCU to fetch the precise, microsecond-stamped MAVLink coordinate. This combined data vector—containing the victim's identifier, the RSSI, and the drone's precise orientation and location—forms the foundational dataset for all subsequent propagation modeling and machine learning operations.

802.11 Probe Request Interception and Promiscuous Sniffing

The core mechanism for detecting survivors relies on the inherent behavior of mobile devices utilizing the IEEE 802.11 (Wi-Fi) standard. To discover available wireless networks, mobile devices perform active scanning by broadcasting unencrypted Layer 2 management frames known as probe requests.

There are two primary variants of these transmissions. A broadcast (or wildcard) probe request features an empty Service Set Identifier (SSID) field, effectively querying any access point within range to announce its presence. Conversely, a directed probe request contains a specific SSID, indicating the device is actively searching for a previously saved network. Because these frames operate pre-association—before any authentication handshake occurs—they are transmitted in plaintext and are visible to any receiving antenna within radio range.

The payload utilizes the onboard WCBN3536A (Qualcomm WCN3980) wireless module, which supports dual-band (2.4 GHz and 5 GHz) 802.11a/b/g/n/ac connectivity. By configuring this network interface card into promiscuous (or monitor) mode using standard Linux networking tools (such as **airmon-ng** or equivalent custom C++ packet sniffers), the system bypasses standard hardware filtering. Instead of dropping packets not destined for its own MAC address, the interface captures all ambient 802.11 traffic in the airspace. A software-level filter is then applied to isolate frames where the subtype identifier matches the hexadecimal value for a probe request (e.g., **0x40**). The resulting capture yields the source MAC address, the signal strength (RSSI), the timestamp, and the payload's Information Elements.

Defeating MAC Randomization via Multi-Dimensional Fingerprinting

Historically, a device's Media Access Control (MAC) address served as a globally unique, permanent identifier. However, recognizing the privacy implications of retail stores and analytics firms passively tracking individuals via probe requests, operating system vendors introduced MAC randomization. Introduced robustly in iOS 14 and Android 10, modern smartphones now generate randomized, locally administered MAC addresses during their scanning phases.

This presents a critical technical hurdle for disaster response; if a single victim's phone continuously rotates its MAC address, standard tracking algorithms will register multiple distinct targets, heavily skewing resource allocation and preventing the accumulation of enough data points to triangulate a single position accurately. To maintain persistent tracking, the payload implements a multi-stage de-randomization and clustering algorithm based on device fingerprinting.

The system exploits the fact that while the MAC address changes, the underlying hardware and the physical characteristics of the transmission do not. The payload analyzes four distinct vectors to cluster randomized MACs back to a single physical device:

First, the system analyzes the Information Elements (IEs) appended to the probe request. IEs contain metadata regarding the device's capabilities, such as supported data rates, Vendor Specific fields, and High Throughput (HT) capabilities. The specific combination, order, and values of these IEs form a highly unique signature tied to the device's specific chipset and firmware. Research has demonstrated that IE fingerprinting alone can reliably track devices across MAC rotations, as the structural blueprint of the packet remains constant.

Second, the algorithm utilizes sequence number tracking. Every 802.11 frame includes a sequence control field that increments linearly with each transmission. Often, the networking stack fails to reset this counter precisely at the moment the MAC address is rotated. If the system detects a MAC address disappearing and a new randomized MAC appearing

instantaneously with a sequentially aligned frame number, a deterministic link is established between the two identifiers.

Third, the payload evaluates temporal burst timing. Devices do not send single probe requests; they transmit rapid bursts across multiple channels in quick succession. The microsecond inter-frame timing within these bursts is governed by proprietary driver implementations and power-management states, creating a unique temporal cadence. If two disparate MAC addresses exhibit an identical rhythm of transmission bursts, they highly correlate to the same device.

Finally, the system analyzes channel scanning patterns. The sequential order in which a device hops through Wi-Fi channels (e.g., channels 1, 6, 11, then 36, 40) is often distinctive. By applying density-based clustering algorithms (such as DBSCAN) or k-means clustering to feature vectors containing these four metrics, the system effectively strips away the anonymity provided by MAC randomization, ensuring that a victim's spatial data is aggregated under a single, persistent internal identifier.

RF Propagation Modeling for Spatial Distance Estimation

Converting a raw RSSI value (measured in dBm) into a reliable physical distance is the most complex variable in wireless localization. Radio waves are subjected to reflection, diffraction, and scattering, severely distorting the signal power over distance. To maximize accuracy, the system does not rely on a single mathematical model. Instead, it dynamically switches between two distinct propagation models based on the environmental topography—utilizing the Log-Distance Path Loss Model for urban rubble and the Two-Ray Ground Reflection Model for flooded terrain.

The Log-Distance Path Loss Model (Collapsed Infrastructure)

In environments characterized by earthquakes or structural collapses, the propagation path is heavily obstructed by concrete, masonry, and steel. These materials possess high signal attenuation properties, rendering standard free-space models entirely inaccurate. For these scenarios, the system employs the Log-Distance Path Loss Model, which predicts propagation loss over a wide range of cluttered environments by incorporating an empirical exponent and a random variable for shadowing.

The total path loss PL(d) in decibels at an arbitrary distance d is defined as:

$$PL(d) = PL(d_0) + 10n\ log_{10}\left(\frac{d}{d_0}\right) + x$$

In this equation, $PL(d_0)$ represents the path loss at a known reference distance d_0 , typically established at 1 meter during pre-deployment calibration. The variable n is the path loss

exponent, which dictates the rate at which the signal decays. In a pure vacuum, n equals 2.0; however, in environments with obstructions, n increases significantly. The variable x is a zero-mean Gaussian distributed random variable with a standard deviation σ , expressed in dB. This variable mathematically models log-normal shadowing—the random attenuation caused by flat fading and obstacles blocking the line of sight.

To implement this model effectively, the system relies on empirical coefficient values established for various environmental densities. The MPU dynamically selects the optimal n and σ parameters based on user-defined inputs regarding the disaster zone's structural density.

Environment Type	Frequency	Path Loss Exponent (n)	Standard Deviation (σ in dB)
Free Space (LOS)	All	2.0	0
Grocery Store	914 MHz	1.8	5.2
Office (Soft Partition)	1.9 GHz	2.6	14.1
Office (Hard Partition)	1.5 GHz	3.0	7.0
Dense Textile/Chemical	4.0 GHz	2.1	9.7

Table 2: Empirical coefficient values for the Log-Distance Path Loss Model across varying structural densities.

By isolating the distance variable d, the algorithm can estimate the radius of the victim relative to the UAV, forming the geometric basis for subsequent localization matrices.

The Two-Ray Ground Reflection Model (Floodplains)

When deploying over large bodies of water, such as those resulting from cyclonic storm surges, the RF propagation dynamics shift radically. Water acts as a highly reflective dielectric surface, fundamentally altering the signal before it reaches the UAV. In this environment, relying on the Log-Distance model yields exponential errors. Instead, the system transitions to the Two-Ray Ground Reflection Model, which accounts for both the direct line-of-sight path and the signal path that reflects off the water's surface.

The received signal is the vector sum of these two overlapping waves. The path difference Δ between the direct ray (d') and the reflected ray (d'') creates a phase difference $\Delta_{\emptyset}$ Using the method of images, and assuming the distance between the transmitter and receiver (d) is significantly larger than the respective antenna heights (h_t and h_r), the phase difference can be approximated using a Taylor series expansion: $\Delta_{\emptyset} \approx \frac{4\pi h_t h_r}{\lambda d}$

For small angles of incidence over highly reflective surfaces, the reflection coefficient Γ approaches -1, meaning the reflected wave is equal in magnitude but 180 degrees out of phase with the direct wave. By retaining the first two terms of the Taylor series, the combined received power P_r simplifies to a closed-form approximation valid in the far-field region: $P_r \approx \frac{P_t G_t G_r (h_t^2 h_r^2)}{d^4}$

This derivation yields a profound insight for UAV deployment: over water, beyond a critical distance, the signal power drops proportionally to the inverse fourth power of the distance d^{-4} equating to a rapid attenuation of 40 dB per decade. This is substantially faster than the 20 dB per decade drop observed in free space. The UAV payload must actively read its MAVLink barometric altitude (h_r) and apply this d^4 attenuation curve; failure to do so would result in the algorithms assuming the victim is drastically further away than they actually are, crippling the rescue effort.

Stochastic State Estimation via Kalman Filtering

Regardless of the precision of the propagation models, raw RSSI readings are intrinsically chaotic. As the UAV navigates the airspace, its own structural components, rotor turbulence, and ambient multipath interference cause rapid, violent fluctuations in signal strength. Utilizing raw, unfiltered RSSI averages directly in distance calculations leads to severe geospatial instability. To solve this, the payload employs a Discrete Kalman Filter to isolate the true signal from the stochastic noise.

The Kalman Filter is a recursive mathematical algorithm designed to estimate the internal state of a linear dynamic system from a series of noisy measurements. It operates continuously across two phases: prediction and update.

During the Prediction Phase, the system projects the current state estimate and error covariance forward in time. Assuming a stationary or slow-moving victim beneath the rubble, the state transition model A approaches an identity matrix.

$$X_k^- = AX_{k-1} + Bu_k$$
$$P_k^- = AP_{k-1}A^T + Q$$

Here, X_k^- represents the a priori state estimate (the predicted RSSI), P_k^- is the a priori error covariance, and Q is the process noise covariance, dictating the inherent uncertainty in the environment.

When the wireless module intercepts a new, noisy RSSI reading (Z_k), the Update Phase begins. The filter computes the Kalman Gain (K_k), a dynamic weighting factor that determines whether to trust the historical prediction or the new measurement.

$$K_k = P_k^- H^T (HP_k^- H^T + R)^{-1}$$

Where H is the observation model and R is the measurement noise covariance, which the system dynamically recalculates based on the standard deviation of recent incoming packets.

The final, smoothed RSSI value (X_k) is then established by blending the prediction and the new measurement via the Kalman Gain: $K_k = X_k^- + K_k(Z_k - HX_k^-)$

Finally, the error covariance (P_k) is updated for the next recursive cycle: $P_k = (1 - K_k H)P_k^-$

By dynamically adjusting the Kalman Gain based on continuous variance analysis, the algorithm effectively suppresses high-frequency noise spikes and drops caused by constructive and destructive wave interference. Empirical studies have demonstrated that applying Gaussian Kalman filtering to Wi-Fi localization reduces position estimation errors by more than 50% in obstructed environments compared to standard temporal averaging.

Machine Learning Paradigms for Geospatial Mapping

Once the RSSI is filtered and mapped to a distance radius, traditional localization relies on trilateration—solving for the intersection of three or more spherical distances (Weighted Non-Linear Least Squares). However, in highly cluttered disaster zones, the geometric assumptions required for exact trilateration frequently break down due to non-line-of-sight (NLOS) conditions. To achieve superior accuracy, the architecture transitions from static geometry to predictive machine learning, leveraging the computational headroom of the Qualcomm QRB2210 processor.

Utilizing TensorFlow and scikit-learn deployed within the Debian Linux environment, the payload executes supervised regression models. The primary algorithm utilized is Random Forest Regression, an ensemble learning method that constructs multiple decision trees during training and outputs the mean prediction of the individual trees. The input feature vector for the model includes the Kalman-filtered RSSI, the MAVLink UAV GPS coordinates, the drone's altitude, and its orientation matrix (pitch, roll, yaw). Random Forests are highly robust against non-linear data distributions and resistant to overfitting, making them exceptionally adept at learning the complex mapping between signal strength variations and physical distance in unpredictable, chaotic environments.

For continuous, real-time trajectory refinement as the UAV flies over a grid, the system can dynamically switch to a deep neural network approach, specifically utilizing a Time-Distributed Autoencoder-Gated Recurrent Unit (TAE-GRU). Unlike static models, recurrent architectures retain an internal memory of previous states. By feeding a time-series sequence of filtered RSSI values and corresponding UAV coordinates into the network, the model analyzes the spatial gradient—understanding whether the signal is strengthening or weakening as the drone moves along a specific vector. This allows the system to converge on the target's precise latitude and longitude with significantly higher accuracy and speed than mathematically rigid trilateration algorithms.

Deployment Feasibility and Evaluation Metrics

The engineering of the Wi-Fi Beacon Locator is heavily informed by the rigorous evaluation criteria established by national innovation bodies, specifically the Smart India Hackathon (SIH) parameters, which emphasize innovation, technical feasibility, scalability, and market readiness.

From an innovation perspective, the solution discards the conventional reliance on costly thermal cameras or acoustic arrays, which fail in specific environmental conditions (e.g., thermal washout in fires, acoustic dampening in floods). By pivoting to RF analysis of ubiquitous mobile devices, the system guarantees a high probability of detection across demographic groups. Technical feasibility is secured by the heterogeneous edge architecture; utilizing a COTS (Commercial Off-The-Shelf) framework like the Arduino Uno Q ensures the solution can be manufactured and deployed rapidly at a fraction of the cost of bespoke military hardware, fulfilling the SIH mandate for cost-effective, scalable solutions.

Furthermore, the system demonstrates high market readiness for integration with state disaster management agencies and the NDRF. During catastrophic events like Super Cyclone Amphan, NDRF teams are tasked with clearing thousands of kilometers of roads and evacuating hundreds of thousands of individuals under extreme time constraints. The ability of this payload to operate entirely independent of localized cellular towers or municipal power grids means it can be deployed at hour zero of a crisis. By providing real-time, precise geospatial coordinates directly to ground rescue teams, the UAV-based Wi-Fi locator dramatically reduces search times, ensuring that critical resources are allocated efficiently to maximize the preservation of human life in the most severe disaster environments.

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